

AI Mock Interview Application

Project Overview

An AI-powered platform designed to help students and job seekers practice technical interviews through dynamically generated questions and performance analytics.

Business Problem

Many candidates lack access to realistic mock interviews and personalized feedback. The system addresses this gap using AI-generated interview scenarios.

Architecture

Frontend built using React.js/Next.js, authentication using Clerk, REST APIs for communication, and Gemini AI API for question generation and feedback.

Modules

User Authentication, Interview Generation, Question Management, Analytics Dashboard, Performance Tracking, Feedback Engine.

Workflow

User logs in, selects interview type, AI generates questions, responses are collected, performance metrics are calculated, and analytics are displayed.

Technical Details

State management, API integration, prompt engineering, responsive UI design, authentication flows, and analytics processing.

Challenges & Solutions

Handling AI response consistency, reducing latency, securing user data, and improving user experience.

Future Enhancements

Voice-based interviews, AI scoring, resume analysis, company-specific interview preparation.

Interview Questions

Explain architecture, Gemini integration, authentication flow, API communication, analytics implementation, and scalability considerations.